

YOLOv11n per-class precision / recall / AP@0.5 by input resolution (clean benchmark, 3 seeds)

Class (cells: P / R / AP@0.5)	640 native	768 infer (1280-tr.)	768 retrain	1024 infer (1280-tr.)	1280 native
Birdnest	0.87 / 0.94 / 0.96	0.93 / 0.95 / 0.97	0.92 / 0.96 / 0.98	0.89 / 0.98 / 0.98	0.89 / 0.96 / 0.98
Self-Exploded Insulator	0.92 / 0.59 / 0.75	0.93 / 0.69 / 0.82	0.93 / 0.67 / 0.80	0.93 / 0.76 / 0.86	0.96 / 0.81 / 0.87
Normal Insulators	0.77 / 0.82 / 0.83	0.81 / 0.80 / 0.84	0.80 / 0.82 / 0.84	0.82 / 0.79 / 0.83	0.84 / 0.80 / 0.84
Normal Damper	0.73 / 0.73 / 0.74	0.87 / 0.75 / 0.80	0.84 / 0.78 / 0.81	0.91 / 0.77 / 0.83	0.89 / 0.79 / 0.81
Broken Insulator	0.69 / 0.60 / 0.67	0.80 / 0.62 / 0.69	0.74 / 0.60 / 0.66	0.83 / 0.60 / 0.70	0.80 / 0.65 / 0.70
Flashover Insulator	0.81 / 0.54 / 0.59	0.79 / 0.51 / 0.58	0.84 / 0.51 / 0.62	0.82 / 0.57 / 0.64	0.81 / 0.58 / 0.66
Defective Damper	0.64 / 0.70 / 0.61	0.58 / 0.62 / 0.59	0.53 / 0.64 / 0.67	0.63 / 0.70 / 0.63	0.66 / 0.61 / 0.62
OVERALL (mAP@0.5)	0.78 / 0.70 / 0.74	0.82 / 0.71 / 0.76	0.80 / 0.71 / 0.77	0.83 / 0.74 / 0.78	0.83 / 0.74 / 0.78

All models trained with the champion recipe (3× defective-damper oversampling + scale-0.9 augmentation, two-stage transfer learning), except 640 native = baseline recipe.
Bold: best AP per row. Deployment point 768 retrain: retraining natively at 768 px recovers the rare-class (Defective Damper) AP that inference-time downscaling loses.